Q_1 = S1: "the tiger sat on the mat"

S2: "the cat sat on the mat"

S3: "the dog sat on the mat"

S4: "the mouse sat on the floor"

1. Using a bigram language Model (LM), calculate the probability of the s1 using smoothing? Training corpus $\{S_2, S_3, S_4\}$.

2. Predict next word after "sat on" using trigram LM on same corpus $\{S_2, S_3, S_4\}$. Show your work.

Q_3 = Assume $\{S_1, S_2\}$ belongs to class 1 and $\{S_3, S_4\}$ to class 2. Train a Naive Bayes classifier. Show all probabilities.

Q_4 = Explain and compare vector representations of documents by BOW (tf-idf) and word embeddings (fastText).